JULY 3, 2026

RECOVERY PROGRESS REPORT

Week 3

This Week's Tasks

- Etch Base-Station PCB
- Pop test
- 3D - Printing shear pins ,Battery Holder, explosion caps
- Kalman Filter Testing
- Make circuit changes for dual memory storage
- Verification of parachute design to ensure they are able to handle the new motor etc
- Changes to base-station Arduino MEGA code
- Fixing BMS charger
- 3

Changes made to Base- station Code

The pins controlling the LED were changed during fixing of the base-station proto-board. We changed the pins in the code base to ensure proper functioning.

Addition of new commands to the base-station:

- Flash erase Command
- Start Simulation Command
- Documentation on how to use the base-station.

N/B unable to etch due to delays in procurement

Pop test

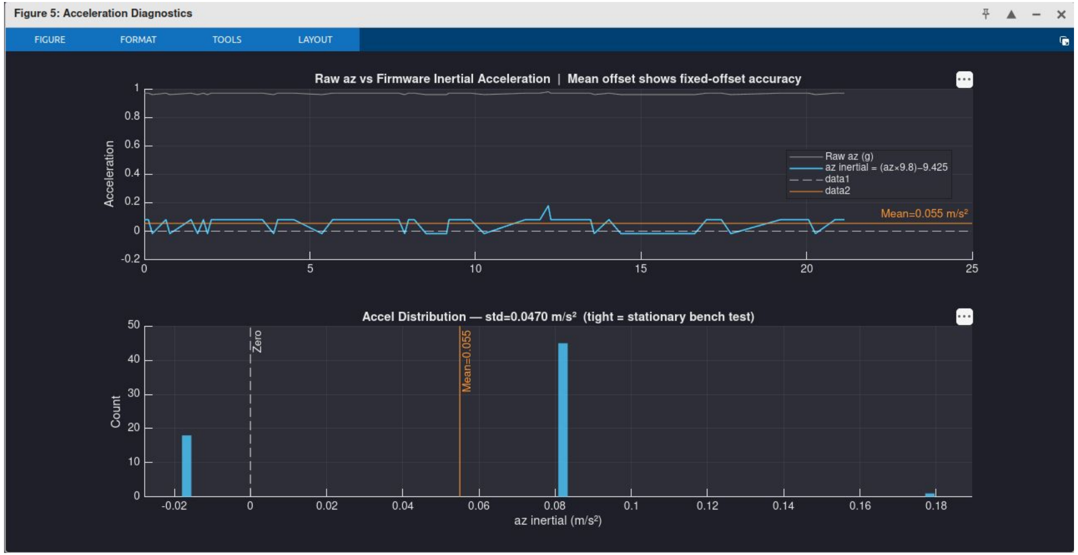
Challenges experienced

Explosion caps have a smaller diameter for the crimson chamber to fit together with the nichrome wires

Inadequate shear pins

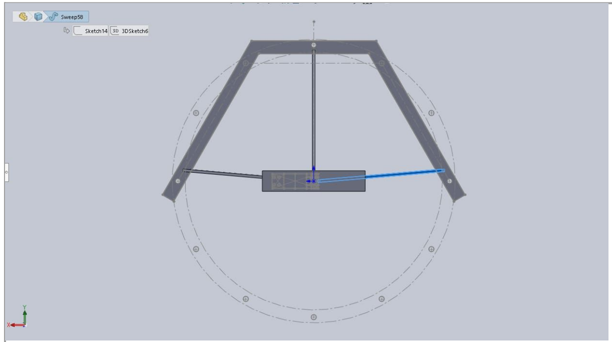
Kalman Filter Fine Tuning

Zero Velocity Update (ZUPT) was incorporated into the Kalman filtering system to reduce velocity drift caused by accelerometer bias and sensor noise during periods when the rocket is stationary.



Finding ways of validating the parachute design

- Static Strength test- Tensile test on shroud lines and the sewn joints of the parachute (to see how much strength the joint can handle without failure)



Before this, a wind tunnel test is to be done to get the coefficient of aerodynamic drag which will be used to obtain our design load that we'll use during the tensile tests

- Deployment test - Should be done using a moving car, preferably a pickup. Replicating one like this.

Next Week’s Tasks

- Etch Base-Station PCB
- 3D printing explosion cap and shear pins
- Changes made to circuit to support Dual memory storage
- Dual pop test
- Parachute extension lines
- Kalman Filter Testing

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